

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – OPEN DATA ANALYSIS PROJECT

Course Code:	DSA05	Course Title:	Query Processing for Data Science in Real Time Applications
CO Assessed:	CO4	Assessment:	Tool 3 – Open Data Analysis Project
Weightage:		Total Marks:	50 Marks
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.. (BL4, BL5 and BL6)			
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Section / Batch:	AI&DS / 24Batch	Date of Submission:	21-08-2026
Faculty In-charge:	K.Veena devi	Assessment Mode:	Individual Submission

Code of Conduct

I __U.Ananya(192424382)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: __U.Ananya_____

Data Collection & Understanding (10 Marks)	Data Cleaning & Preparation (10 Marks)	Exploratory Data Analysis (10 Marks)	Visualization & Dashboard Design (10 Marks)	Demonstration (10 Marks)	Total (50 Marks)

1. Global COVID-19 Trends and Vaccination Analysis

1. Introduction

COVID-19 affected countries worldwide and caused major changes in infection rates, deaths, and vaccination levels. This project analyzes COVID-19 data to identify trends across countries and understand the relationship between vaccination, cases, and mortality. Python is used for data cleaning, analysis, visualization, and statistical comparison.

2. Objectives

- To clean and preprocess COVID-19 data.
- To handle missing and duplicate values.
- To analyze COVID-19 cases and deaths across countries.
- To compare vaccination coverage.
- To identify trends over time.
- To calculate mortality and vaccination rates.
- To detect unusual values using outlier analysis.
- To study relationships using correlation analysis.
- To visualize results using charts and graphs.
- To provide useful public-health recommendations.

3. Dataset

The dataset contains:

- Date
- Country
- Population
- Total Cases
- New Cases
- Total Deaths
- New Deaths
- People Vaccinated
- People Fully Vaccinated

4. Python Code

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("covid19_global_vaccination.csv")

print(df.head())

print(df.info())

print(df.isnull().sum())

df = df.drop_duplicates()

df["Date"] = pd.to_datetime(df["Date"], errors="coerce")

num_cols = [

    "Population", "Total_Cases", "New_Cases",

    "Total_Deaths", "New_Deaths",

    "People_Vaccinated", "People_Fully_Vaccinated"

]

for col in num_cols:

    df[col] = pd.to_numeric(df[col], errors="coerce")

    df[col] = df[col].fillna(0)

df["Mortality_Rate"] = np.where(

    df["Total_Cases"] > 0,

    (df["Total_Deaths"] / df["Total_Cases"]) * 100,
```

0

)

```
df["Vaccination_Rate"] = (
```

```
    df["People_Vaccinated"] /
```

```
    df["Population"]
```

```
) * 100
```

```
latest = (
```

```
    df.sort_values("Date")
```

```
    .groupby("Country")
```

```
    .tail(1)
```

```
)
```

```
print("\nCountry-wise Results:")
```

```
print(
```

```
    latest[
```

```
        ["Country", "Total_Cases",
```

```
        "Total_Deaths", "Vaccination_Rate",
```

```
        "Mortality_Rate"]
```

```
    ]
```

```
)
```

```
plt.figure(figsize=(10,5))
```

```
sns.barplot(
```

```
    data=latest,
```

```
x="Country",  
  
y="Total_Cases"  
  
)  
  
plt.title("COVID-19 Cases by Country")  
  
plt.xticks(rotation=45)  
  
plt.tight_layout()  
  
plt.show()
```

```
plt.figure(figsize=(10,5))
```

```
sns.barplot(  
  
    data=latest,  
  
    x="Country",  
  
    y="Total_Deaths"  
  
)
```

```
plt.title("COVID-19 Deaths by Country")  
  
plt.xticks(rotation=45)  
  
plt.tight_layout()  
  
plt.show()
```

```
plt.figure(figsize=(10,5))
```

```
sns.barplot(  
  
    data=latest,  
  
    x="Country",  
  
    y="Vaccination_Rate"
```

)

```
plt.title("Vaccination Rate by Country")
```

```
plt.ylabel("Vaccination Rate (%)")
```

```
plt.xticks(rotation=45)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
plt.figure(figsize=(12,6))
```

```
for country in df["Country"].unique():
```

```
    data = df[df["Country"] == country]
```

```
    plt.plot(
```

```
        data["Date"],
```

```
        data["Total_Cases"],
```

```
        marker="o",
```

```
        label=country
```

```
    )
```

```
plt.title("COVID-19 Cases Over Time")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Total Cases")
```

```
plt.legend()
```

```
plt.xticks(rotation=45)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
cols = [  
  
    "Total_Cases",  
  
    "Total_Deaths",  
  
    "People_Vaccinated",  
  
    "People_Fully_Vaccinated",  
  
    "Mortality_Rate",  
  
    "Vaccination_Rate"  
  
]  
  
corr = df[cols].corr()  
  
print("\nCorrelation:")  
  
print(corr)  
  
plt.figure(figsize=(9,6))  
  
sns.heatmap(  
  
    corr,  
  
    annot=True,  
  
    cmap="coolwarm"  
  
)  
  
plt.title("COVID-19 Correlation Heatmap")  
  
plt.tight_layout()  
  
plt.show()  
  
Q1 = df["Total_Cases"].quantile(0.25)  
  
Q3 = df["Total_Cases"].quantile(0.75)
```

$IQR = Q3 - Q1$

```
outliers = df[
```

```
(df["Total_Cases"] < Q1 - 1.5 * IQR) |
```

```
(df["Total_Cases"] > Q3 + 1.5 * IQR)
```

```
]
```

```
print("\nOutliers:")
```

```
print(outliers[["Country", "Date", "Total_Cases"]])
```

Output:

```
...
      Date Country Population Total_Cases Total_Deaths \
0  2020-03-01  India  1380004385          3           0
1  2020-04-01  India  1380004385        1998          58
2  2020-06-01  India  1380004385       190535        5394
3  2021-01-01  India  1380004385      1026674       148994
4  2021-06-01  India  1380004385      3036284       398484

      People_Vaccinated  New_Cases  New_Deaths  People_Fully_Vaccinated
0                   0         3.0         0.0                   0.0
1                   0        1995.0        58.0                   0.0
2                   0       188537.0       5336.0                   0.0
3                   0      10076139.0      143600.0                   0.0
4                220000000      20096174.0      249490.0       143000000.0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35 entries, 0 to 34
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                35 non-null    object
1   Country                            35 non-null    object
2   Population                          35 non-null    int64
3   Total_Cases                        35 non-null    int64
4   Total_Deaths                       35 non-null    int64
5   People_Vaccinated                  35 non-null    int64
6   New_Cases                          35 non-null    float64
7   New_Deaths                         35 non-null    float64
8   People_Fully_Vaccinated            35 non-null    float64
```



```

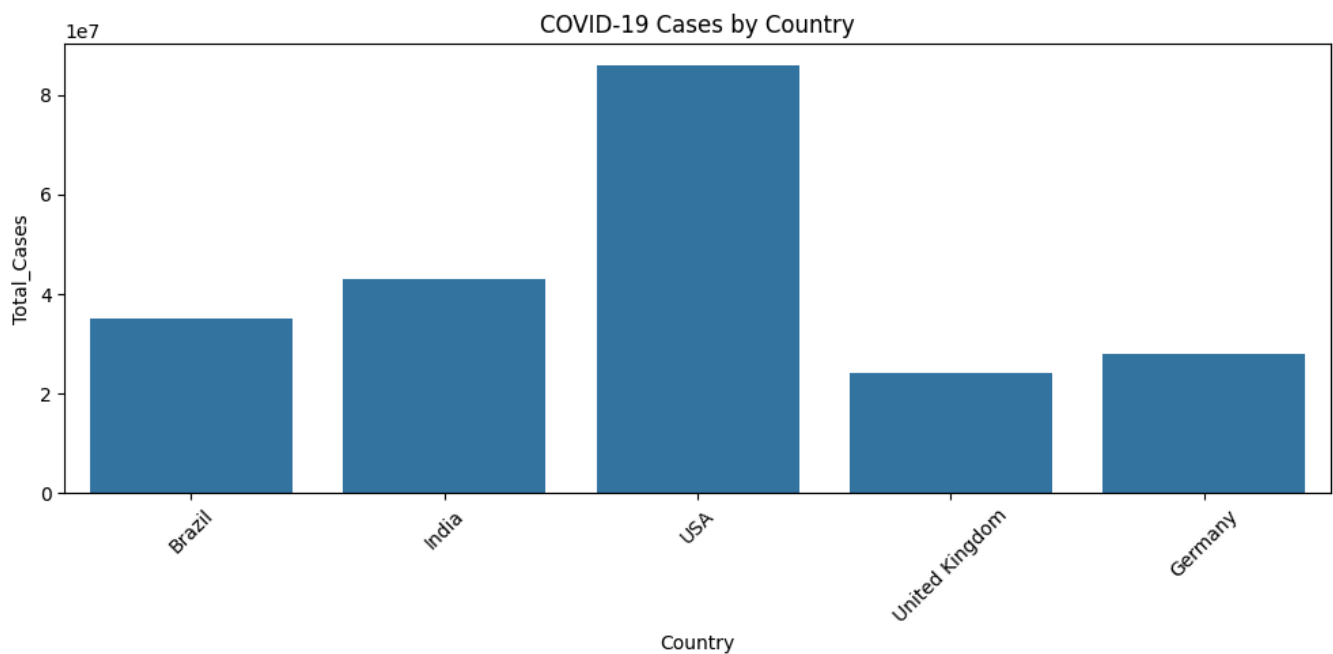
... dtypes: float64(3), int64(4), object(2)
memory usage: 2.6+ KB
None
Date          0
Country       0
Population    0
Total_Cases   0
Total_Deaths  0
People_Vaccinated  0
New_Cases     0
New_Deaths    0
People_Fully_Vaccinated  0
dtype: int64

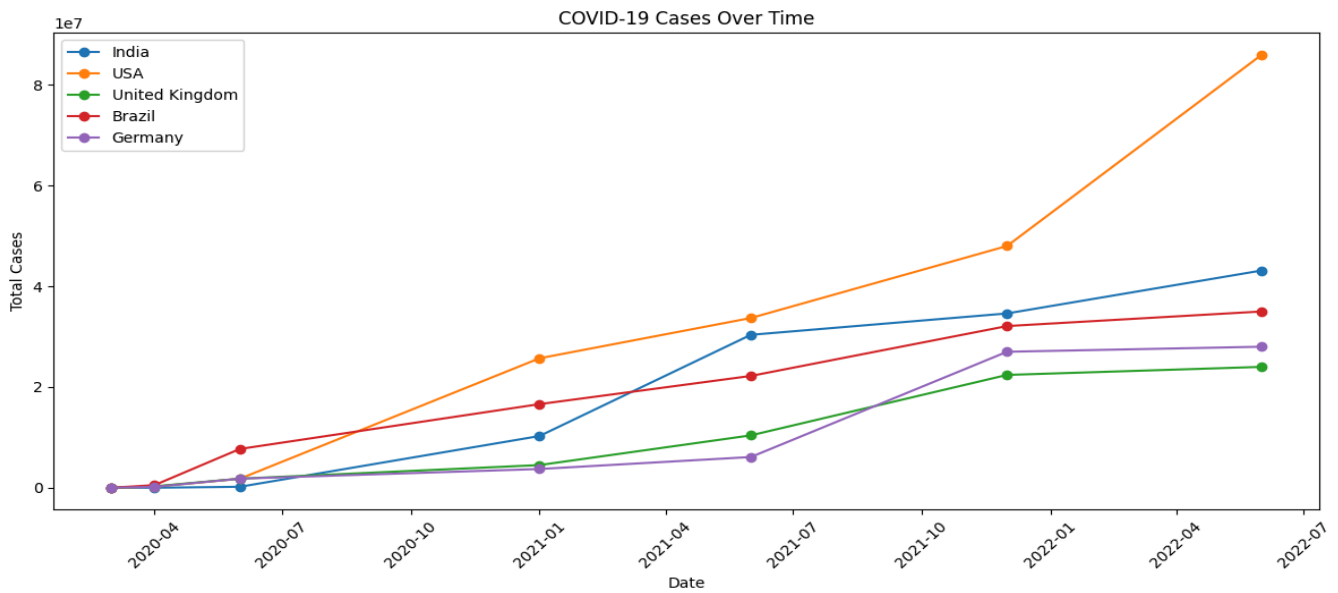
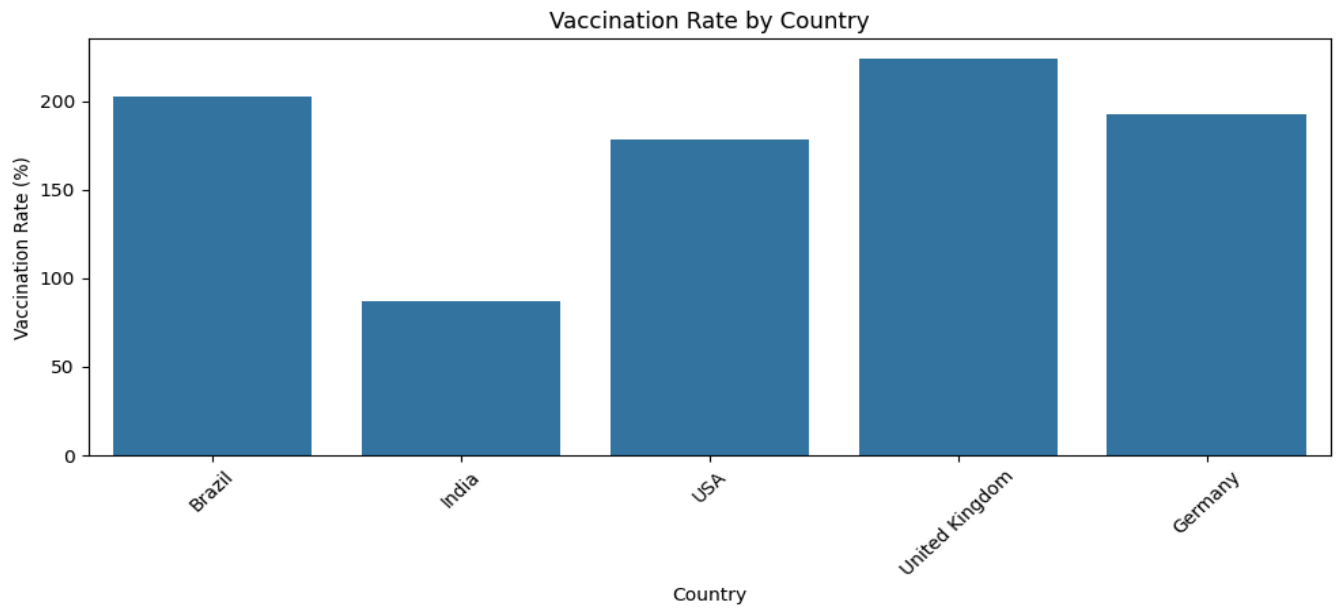
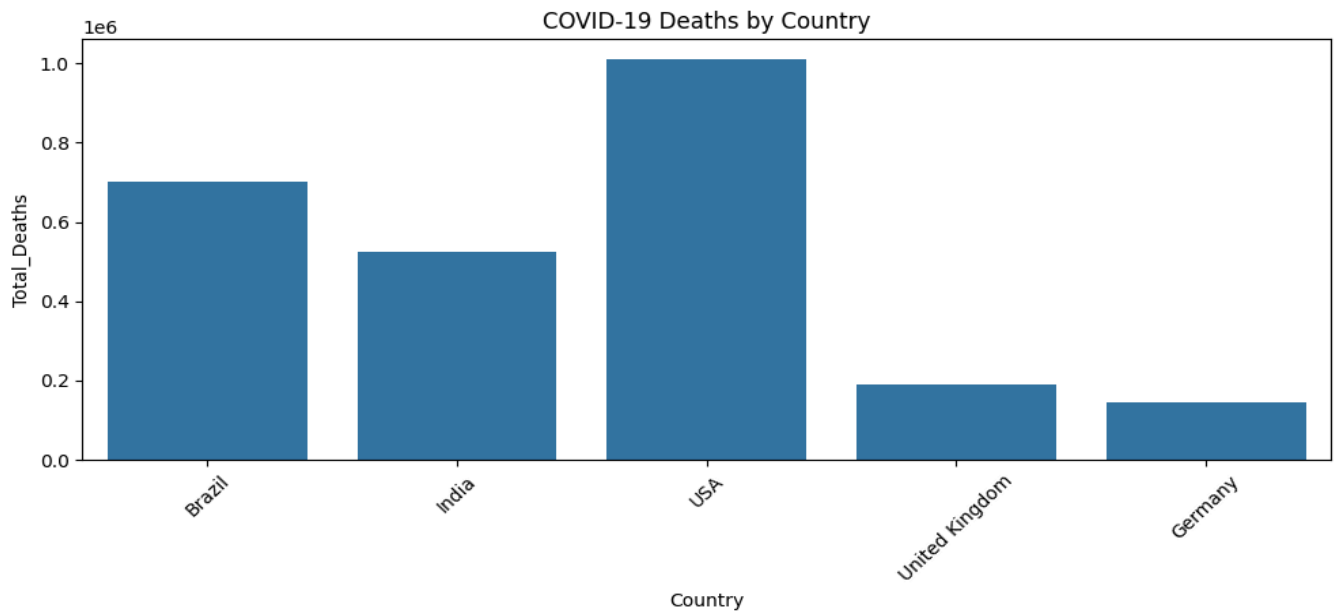
Country-wise Results:

```

	Country	Total_Cases	Total_Deaths	Vaccination_Rate \
27	Brazil	35000000	700000	202.830189
6	India	43139294	524954	86.956245
13	USA	86000000	1010000	178.247734
20	United Kingdom	24000000	190000	223.880597
34	Germany	28000000	145000	192.771084

	Mortality_Rate
27	2.000000
6	1.216881
13	1.174419
20	0.791667
34	0.517857





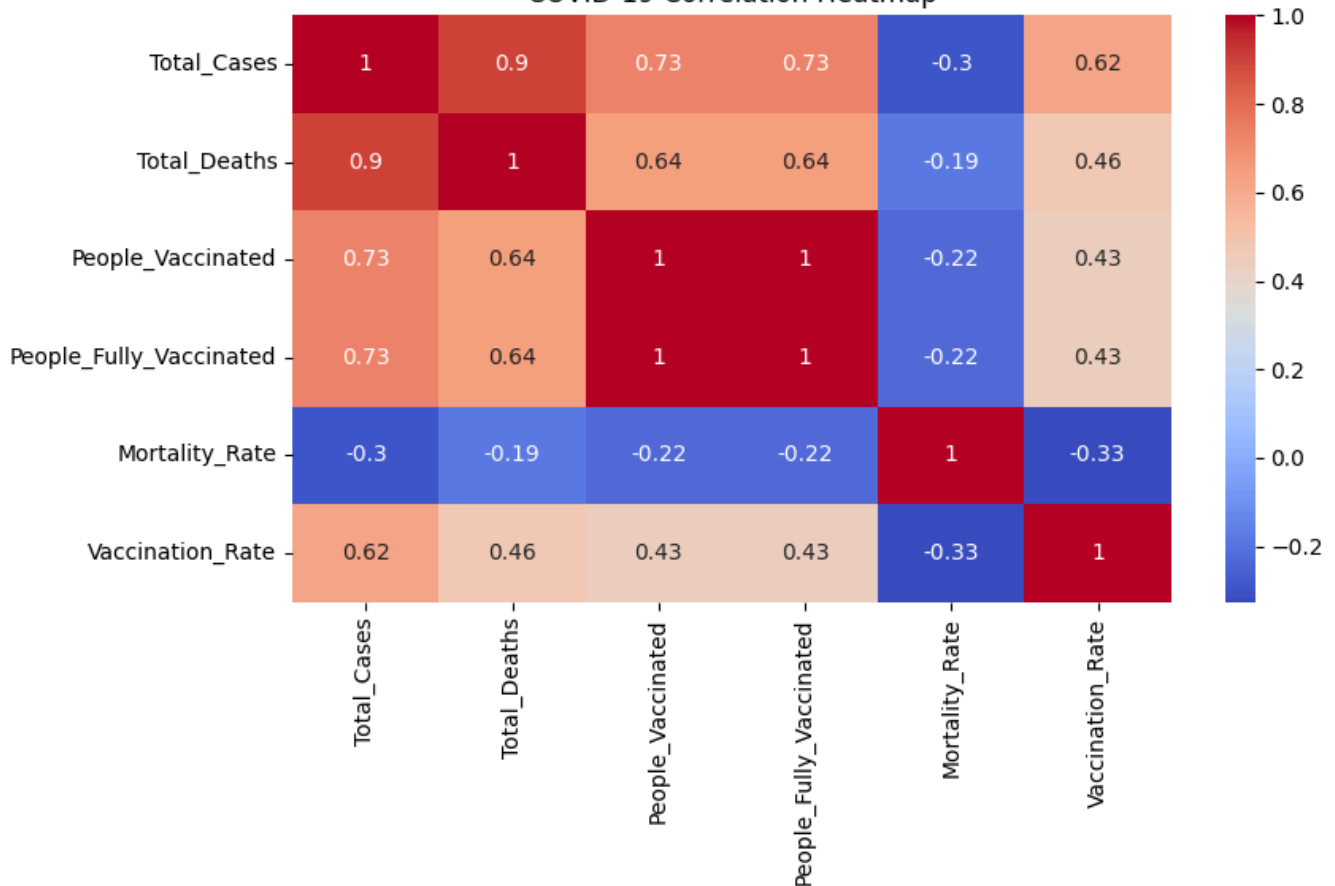
Correlation:

	Total_Cases	Total_Deaths	People_Vaccinated	\
Total_Cases	1.000000	0.903111	0.726575	
Total_Deaths	0.903111	1.000000	0.644613	
People_Vaccinated	0.726575	0.644613	1.000000	
People_Fully_Vaccinated	0.726575	0.644613	1.000000	
Mortality_Rate	-0.299865	-0.186676	-0.222894	
Vaccination_Rate	0.624983	0.464612	0.433543	

	People_Fully_Vaccinated	Mortality_Rate	\
Total_Cases	0.726575	-0.299865	
Total_Deaths	0.644613	-0.186676	
People_Vaccinated	1.000000	-0.222894	
People_Fully_Vaccinated	1.000000	-0.222894	
Mortality_Rate	-0.222894	1.000000	
Vaccination_Rate	0.433543	-0.325977	

	Vaccination_Rate
Total_Cases	0.624983
Total_Deaths	0.464612
People_Vaccinated	0.433543
People_Fully_Vaccinated	0.433543
Mortality_Rate	-0.325977
Vaccination_Rate	1.000000

COVID-19 Correlation Heatmap



Outliers:

	Country	Date	Total_Cases
13	USA	2022-06-01	86000000

5. Findings

The analysis helps identify countries with high numbers of cases and deaths and compare their vaccination coverage. The time-series graph shows how cases changed during different periods. Correlation analysis shows relationships between COVID-19 and vaccination variables, while outlier detection identifies unusually high case values.

6. Recommendations

- Continue monitoring COVID-19 trends.
- Improve the quality of health data.
- Monitor vaccination coverage.
- Provide timely public-health information.
- Focus healthcare resources on high-risk populations.
- Use dashboards for continuous monitoring.

7. Conclusion

This project demonstrates how Python, data cleaning, statistical analysis, and visualization can be used to understand global COVID-19 trends. The analysis provides useful information about cases, deaths, vaccination coverage, and their relationships, helping support better public-health decision-making.